GUM Gen Form

$$\Rightarrow \quad Y = f(X_1, X_2 \ldots X_N) = \boxed{C_{\text{final}}}$$

$$\Rightarrow \quad u_c^2(y) = \sum_{i=1}^{N} c_i^2 u^2(x_i) + 2 \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} c_i c_j u(x_i, x_j)$$

$$\Rightarrow \quad c_i = \frac{\partial f}{\partial x_i}$$

Inputs

Fixed Inputs ($u=0$)

C_0 = Starting OD

C_{target} = target OD

$\tilde{V}_{transfer}$ = volume intended in initial split

"~" means ideal volume

$\tilde{v}_{i,j}$; $\tilde{v}_{d,k}$ = nominal dispense volumes max set by tip size

Uncertain inputs

$\varepsilon_{sys,t}$ = Systemic transfer bias

$\varepsilon_{sys,d}$ = Systemic dilution bias

$\varepsilon_{rand,t,j}$ = Random transfer error at dispense j

$\varepsilon_{rand,d,k}$ = Random dilution error of dispense k

Dispense error

no tilde is true value

$$v_j = \tilde{v}_j(1 + \varepsilon_{sys} + \varepsilon_{rand,j})$$

$\bar{V}_{step}$ = total volume of step to be dispensed over m dispenses

$$= \sum \tilde{v}_j(1 + \varepsilon_{sys} + \varepsilon_{rand,j}) = \tilde{V}_{step}(1 + \varepsilon_{sys}) + \sum_j \tilde{v}_j \varepsilon_{rand}$$

Let $\bar{V}_{step} = \tilde{\bar{V}}_{step}(1 + \varepsilon_{total})$,

$$\tilde{\bar{V}}_{step}(1 + \varepsilon_{total}) = \tilde{\bar{V}}_{step} + \tilde{\bar{V}}_{step}\varepsilon_{sys} + \sum_j \tilde{v}\,\varepsilon_{rand}$$

$\div$

$\Rightarrow$

$$\boxed{\varepsilon_{total} = \varepsilon_{sys} + \sum_j \frac{\tilde{v}}{\tilde{\bar{V}}_{step}}\varepsilon_{rand} \qquad \cdot \; (e) = \frac{\tilde{v}}{\tilde{\bar{V}}}}$$

step error weight

Concentration - Volume Gen form

$$C_1 V_1 = C_2 V_2$$

Step Balance

$$C_0 \tilde{V}_{transfer} = C_{target} V_{final} = C_{target} (\tilde{V}_{transfer} + \tilde{V}_d)$$

$$\Rightarrow \boxed{\tilde{V}_d = \tilde{V}_{transfer} \left(\frac{C_0}{C_{target}} - 1 \right)}$$

Let, $V_{transfer} = \tilde{V}_{transfer} (1 + \varepsilon_{total,t})$

Let, $V_d = \tilde{V}_d (1 + \varepsilon_{total,d})$

Culture Mass Balance

from previous balance,

$$f(\cdot) = C_{final} = \frac{C_0 V_{transfer}}{V_{transfer} + V_d} = \frac{C_0 \tilde{V}_{transfer}(1+\varepsilon_{total,t})}{\tilde{V}_t(1+\varepsilon_{total,t}) + \tilde{V}_d(1+\varepsilon_{total,d})}$$

Input Target Mass balance

$$C_0 \tilde{V}_{transfer} = C_{target} \tilde{V}_{final} = C_{target}(\tilde{V}_{transfer} + \tilde{V}_{dispense})$$

$$\Rightarrow C_{target} = \frac{C_0 \tilde{V}_{transfer}}{\tilde{V}_{transfer} + \hat{V}_d} = \frac{C_{target}}{C_0} = \frac{\tilde{V}_t}{\tilde{V}_t + \hat{V}_d}$$

Culture Share

$$\frac{\tilde{V}_{transfer}}{\tilde{V}_{transfer} + \hat{V}_d} + \frac{\tilde{V}_d}{\tilde{V}_{transfer} + \hat{V}_d} = 1$$

$$\tilde{V}_{transfer} = \tilde{V}_t$$

Quotient rule

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

Solve uncertainty → want ∂(final)/∂ ε_{total} for t & d

Step 1: transfer

$$C_{final} = \frac{C_0 V_{transfer}}{V_{transfer} + V_d}$$

f ↶ g ↷

$$\begin{cases} f' = C_0 \\ g' = 1 + 0 = 1 \end{cases}$$

$$\frac{\partial C_{final}}{\partial V_t} = \frac{C_0(V_{transfer} + V_d) - C_0 V_{transfer}(1)}{(V_t + V_d)^2}$$

$$= \frac{C_0 V_t + C_0 V_d - C_0 V_t}{(V_t + V_d)^2} = \frac{C_0 V_d}{(V_t + V_d)^2}$$

$$\hookrightarrow \bar{v}_{transfer} = \tilde{v}_{transfer}(1 + \varepsilon_{total,t}) = \tilde{v}_t + \tilde{v}_t \varepsilon_{total,t}$$

$$\frac{\partial v_t}{\partial \varepsilon_{total,t}} = 0 + \tilde{v}_t = \tilde{v}_t$$

$$\Rightarrow \frac{\partial C_{final}}{\partial \varepsilon_{total,t}} = \frac{\partial C_{final}}{\partial v_t} \cdot \frac{\partial v_t}{\partial \varepsilon_{total,t}} = \frac{C_0 v_d \cdot \tilde{v}_t}{(v_t + v_d)^2}$$

Step 2: Diviation

$\sim$ held constant so factor out of ∂v_d

$$\rightarrow C_{final} = \frac{C_0 v_t}{v_t + v_d} = C_0 v_t (v_t + v_d)^{-1}$$

$$\frac{\partial C_{final}}{\partial v_d} = C_0 v_t \cdot -(v_t + v_d)^{-2} = -\frac{C_0 v_t}{(v_t + v_d)^2}$$

$$\rightarrow \quad V_d = \tilde{V}_d \left(1 + \varepsilon_{total,d}\right) = \tilde{V}_d + \tilde{V}_d\, \varepsilon_{total,d}$$

$$\frac{\partial V_d}{\partial \varepsilon_{total,d}} = \tilde{V}_d$$

$$\Rightarrow \quad \frac{\partial C_{final}}{\partial \varepsilon_{total,d}} = -\frac{C_0 V_t \tilde{V}_d}{(V_t + V_d)^2}$$

for uncertainty budget, find $\partial C_{final} / \partial \varepsilon_{total}$ w/ non error dependent values by using identity at $\varepsilon_{total} = 0$

$$V_t = \tilde{V}_t \left(1 + \varepsilon_{total,t}\right) = \tilde{V}_t (1 + 0) = \tilde{V}_t$$

$$V_d = \tilde{V}_d \left(1 + \varepsilon_{total,d}\right) = \tilde{V}_d$$

$$\Rightarrow \frac{\partial C_{final}}{\partial \varepsilon_{total,t}} = \frac{C_0 \tilde{V}_t \tilde{V}_d}{(\tilde{V}_t + \tilde{V}_d)^2} \quad \& \quad \frac{\partial C_{final}}{\partial \varepsilon_{total,d}} = -\frac{C_0 \tilde{V}_t \tilde{V}_d}{(\tilde{V}_t + \tilde{V}_d)^2}$$

in terms of C

$$\Rightarrow \frac{\partial C_{final}}{\partial \varepsilon_{total,t}} = C_0 \frac{\tilde{V}_t}{(\tilde{V}_t+\tilde{V}_d)} \cdot \frac{\tilde{V}_d}{(\tilde{V}_t+\tilde{V}_d)} = C_0 \frac{\tilde{V}_t}{(\tilde{V}_t+\tilde{V}_d)}\left(1-\frac{\tilde{V}_t}{\tilde{V}_t+\tilde{V}_d}\right)$$

$$= C_0 \cdot \frac{C_{target}}{C_0} \cdot \left(1-\frac{C_{target}}{C_0}\right) = C_{target}\left(1-\frac{C_{target}}{C_0}\right)$$

$$\frac{\partial C_{final}}{\partial \varepsilon_{total,d}} = -C_{target}\left(1-\frac{C_{target}}{C_0}\right)$$

Per-Output

$$\varepsilon_{total,t} = \varepsilon_{sys,t} + \sum_j \omega_j \varepsilon_{rand,j,t}$$

$$\frac{\partial \varepsilon_{total,t}}{\partial \varepsilon_{sys,t}} = 1 \qquad j \qquad \frac{\partial \varepsilon_{total,t}}{\partial \varepsilon_{rand,j,t}} = \omega_j$$

$$\varepsilon_{total,d} = \varepsilon_{sys,d} + \sum_k \omega_k \varepsilon_{rand,d}$$

$$\frac{\partial \varepsilon_{total,d}}{\partial \varepsilon_{sys,d}} = 1 \qquad \therefore \quad \frac{\partial \varepsilon_{total,d}}{\partial \varepsilon_{rand,d}} = \omega_k$$

C_i , Sensitivity Coeff

$$\frac{\partial C_{final}}{\partial \varepsilon_{sys,d}} = -C_{target}\left(1 - C_{target}/C_0\right)$$

$$\frac{\partial C_{final}}{\partial \varepsilon_{rand,d}} = -\omega_k C_{target}\left(1 - C_{target}/C_0\right)$$

$$\frac{\partial C_{final}}{\partial \varepsilon_{sys,t}} = C_{target}\left(1 - C_{target}/C_0\right)$$

$$\frac{\partial C_{final}}{\partial \varepsilon_{sys,t}} = C_0 \, C_{target} \left(1 - C_{target}/C_d\right)$$

Error Model

$$Y - Y_0 = \sum_i c_i \delta_i$$

$$C_{final} - C_{target} \approx \sum_{i=1}^{N} c_i \delta_i$$

Covariances

ε_{rand} vs ε_{rand} = independent dispersers

ε_{sys} v ε_{rand} = no covariance b/c rand is scatter & sys is offset

$\varepsilon_{sys,t}$ vs $\varepsilon_{sys,d}$ = $r\, u(\varepsilon_{sys,t})\, u(\varepsilon_{sys,d})$

↳ $r = 1$ b/c same pipette

$$\Rightarrow u_c^2(C_{final}) = \sum_j c_{t,j}^2 u^2(\varepsilon_{rand,t}) + \sum_k c_{d,k}^2 u^2(\varepsilon_{rand,d,k})$$

$$+ c_{sys,t}^2 u^2(\varepsilon_{sys,t}) + c_{sys,d}^2 u^2(\varepsilon_{sys,d})$$

$$+ 2 c_{sys,t} c_{sys,d} r\, u(\varepsilon_{sys}) u(\varepsilon_{sys,d})$$

at $r = 1$,

$$u_c = C_{target}\left(1 - \frac{C_{target}}{C_0}\right) u(\varepsilon_{rand}) \sqrt{\sum_j \omega_{t,j}^2 + \sum \omega_{d,k}^2}$$